

CATS⁺

Competishun Advance

Training Sessions

(Lecture - XIV)
(Matrix Special)

Concepts

C-1 If ~~rows~~^{columns} of a determinant are in A.P. (different) then its value = 0.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 10 & 15 \\ 7 & 10 & 13 \end{bmatrix} \Rightarrow |A| = 0.$$

C-2 E.R.T.

$$\begin{bmatrix} 1 & 2 & 3 \\ 5 & -1 & 6 \\ 8 & 7 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 9/11 \\ 0 & 7 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 5/11 & -1/11 & 0 \\ 8 & 0 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot$$

C-3.

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 + a_2 + a_3 & a_1 + a_2 & a_1 \\ b_1 + b_2 + b_3 & b_1 + b_2 & b_1 \\ c_1 + c_2 + c_3 & c_1 + c_2 & c_1 \end{bmatrix} \begin{matrix} a_2 \\ b_2 \\ c_2 \end{matrix}$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$= \begin{bmatrix} c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \\ 2a_1 & 2a_2 & 2a_3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} 0 & - & - \\ 0 & - & - \\ 1 & - & - \end{bmatrix}$$

$$= \begin{bmatrix} b_1 & b_2 & b_3 \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} 0 & - & - \\ 0 & - & - \\ 1 & - & - \end{bmatrix}$$

C-4

MCQ's

↓
A
B
C
D ↓

Que

For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

$$|E| = 0 \\ = |F|$$

If Q is a nonsingular matrix of order 3×3 , then which of the following statements is (are) **TRUE** ?

(A) ~~$F = PEP$~~ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) ~~$|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$~~

(C) ~~$|(EF)^3| > |EF|^2$~~

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

$$P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$P = P^{-1}$$

$$F = PEP \\ \Rightarrow |F| = |P| \cdot |E| \cdot |P| \\ = 0$$

$$\begin{aligned} P^{-1}EP + F &= PEP + F = F + F = 2F \\ E + P^{-1}FP &= E + PFP = 2E \end{aligned}$$

$$\textcircled{B} \quad |\varepsilon Q + PFQ^{-1}| = |\underline{\varepsilon} Q + \underline{\varepsilon} P Q^{-1}|$$

$$= |\underline{\varepsilon} (Q + P Q^{-1})|$$

$$= \cancel{|\underline{\varepsilon}|} \cdot |Q + P Q^{-1}|$$

$$\begin{aligned} P. \quad & f = P \varepsilon P \rightarrow I \\ & \underline{P} f = \cancel{P^2} \varepsilon P \\ & = \underline{\varepsilon} P. \end{aligned}$$

$$\begin{aligned} F &= P \varepsilon P \\ P F P &= \cancel{P^2} \varepsilon \cancel{P^2} \\ &\rightarrow I \\ &= \varepsilon \end{aligned}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}$$

$$P\Sigma P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix} \cdot P$$

$$R_2 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 8 & 13 & 18 \\ 2 & 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 8 & 13 & 4 \\ 2 & 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

One

For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) TRUE ?

- ☒ (A) $|FE| = |I - FE| |FGE|$
- ☒ (C) $\underline{EFG} = \underline{GEF}$
- ☒ (B) $(I - FE)(I + FGE) = I$
- ☒ (D) $(I - FE)(I - FGE) = I$

$G = (I - EF)^{-1}$

$(I - EF) \cdot G = I$ $G \cdot (I - EF) = I$

$G - EFG = I$ $G - GEF = I$

$\boxed{EFG = GEF}$ $G - I = \underline{GEF}$

(A) $|FE| = |I - FE| \cdot |FGE|$

$= |FGI - FEFGI|$

$= |FGI - F(G - I)E|$

$= |FGI - FGI + FE| = |FE|$

①

$$\begin{aligned}
 & (I - f\varepsilon)(I - fG\varepsilon) \\
 &= I - \underline{f\varepsilon} - \cancel{fG\varepsilon} + \underbrace{f\varepsilon fG\varepsilon}_{(G-I)} \\
 &= \cancel{fG\varepsilon} - \underline{f\varepsilon} \\
 &= \underline{I - 2f\varepsilon}
 \end{aligned}$$

②

$$\begin{aligned}
 & (I - f\varepsilon)(I + fG\varepsilon) = I \\
 &= I - f\varepsilon + fG\varepsilon - \underbrace{f\varepsilon fG\varepsilon}_{G-I} \\
 &= I - \cancel{f\varepsilon} + \cancel{fG\varepsilon} - f(G-I)\varepsilon \\
 &= \cancel{-fG\varepsilon} + \cancel{f\varepsilon} \\
 &= I
 \end{aligned}$$

Ques

Let $x \in \mathbb{R}$ and let

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \text{ and } R = PQP^{-1}.$$

Then which of the following is/are correct

~~(1) there exists a real number x such that $PQ = QP$~~

~~(2) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$ for all $x \in \mathbb{R}$~~

$$\Rightarrow 4(10 - x^2) = 40 - x^2$$

~~(3) For $x = 1$ there exists a unit vector $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ for which are $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$~~

~~(4) For $x = 0$ if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$ then $a + b = 5$~~

$$\Rightarrow |R| = |P| \cdot |Q| \cdot |P^{-1}|$$

$$= |Q| = 4(12 - x^2) = 48 - x^2$$

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{\kappa=0}$$

$$R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1/2 & -1/3 \\ 0 & 0 & 1/3 \end{bmatrix}$$

$$= PQP^{-1}$$

$$\begin{bmatrix} 2 & 1 & 2/3 \\ 0 & 4 & 4/3 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$$

$$a + \frac{2}{3}b = 4$$

$$2 + a + \frac{2}{3}b = 6$$

$$4a + \frac{4}{3}b = 6a$$

$$\frac{2}{3}b + \frac{2}{3}b = 4 \Rightarrow \boxed{b=3}$$

$$\boxed{a=2}$$

$$a = \frac{2}{3}b$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1/2 & 0 \\ 0 & 1/2 & -1/3 \\ 0 & 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & -4/3 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2/3 \\ 0 & 4 & 4/3 \\ 0 & 0 & 6 \end{bmatrix}$$

$$-\frac{4}{3} + \frac{2}{3} - \frac{8}{3}$$

$$PQ = QP$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & 8 & 12 \\ 0 & 0 & 18 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 8 & 8 \\ 0 & 0 & 18 \end{bmatrix}$$

Ques

Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj } M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$

$$\text{adj}(A) = |A| \cdot A^{-1}$$

$$A \cdot \text{adj}(A) = |A| I_n$$

where a and b are real numbers. Which of the following options is/are correct?

$$|\text{adj}(A)| = |A|^{n-1}$$

~~(1) $\det(\text{adj } M^2) = 81$~~

$$|\text{adj}(M^2)| = |M^2|^2 = |M|^4$$

~~(2) $a + b = 3$~~

~~(3) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$~~

~~(4) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$~~

✂

$$M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$$

$$\text{adj}(M) = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$$

$$M \cdot \text{adj}(M) = |M| \cdot I_n = \begin{bmatrix} |M| & 0 & 0 \\ 0 & |M| & 0 \\ 0 & 0 & |M| \end{bmatrix}$$
$$= \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$$
$$\begin{bmatrix} -1 & 2 & -1 \\ -12 & 9 & -12 \end{bmatrix} = |M| I_3 \Rightarrow |M| = -2$$

③

$$M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = M^{-1} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \frac{\text{adj}(M)}{|M|} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -4 & 3 & -1 \\ \frac{5}{2} & -3\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\alpha = 1$$

$$\beta = -1$$

$$\gamma = 1$$

$$\alpha - \beta + \gamma$$

$$\stackrel{\text{---}}{=} \textcircled{3}$$

$$\begin{array}{r} -4 \\ -6 \\ -3 \\ \frac{1}{2} - 3\frac{1}{2} \end{array}$$

④

$$(\text{adj}(M))^{-1} + \text{adj}(M^{-1})$$

$$= (|M| M^{-1})^{-1} + |M^{-1}| \cdot (M^{-1})^{-1}$$

$$= \frac{M}{|M|} + \frac{M}{|M|} = \frac{2M}{|M|}$$

$$= \frac{2M}{-2} = -M$$

$$(kA)^{-1} = \frac{A^{-1}}{k}$$

$$8 - 5a = |M| = -2$$

$$\boxed{a = 2}$$

$$a_{33} = -4 + 2b = -2$$

$$\boxed{b = 1}$$